EMBODIED ATTENTION BUDGETING: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

Anonymous authors
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ABSTRACT

Foundation-model policies often allocate attention according to visual or linguistic token salience. The hypothesis tested here is that robot policies should instead allocate perception and action attention by physical consequence: contacts, hazards, occlusions, and action-critical regions that change the outcome should receive budget even when visually unremarkable. We rebuilt this generated paper around a deterministic local attention-budget benchmark with four manipulation tasks, five distribution shifts, eight methods, seven seeds, 47,040 main rollouts, 8,232 ablation rollouts, and 141,120 stress rollouts. The result is promising but not submission-ready. On the combined hard-shift split, embodied consequence budgeting reaches 0.693 task success, 0.562 critical-event recall, and 0.062 wasted attention. However, risk-aware attention reaches 0.676 task success and the paired gain is only 0.017 ± 0.041. An ablation without the action-critical contact map reaches 0.705 success, and at maximum combined stress risk-aware attention beats the proposed method. We therefore mark this paper **KILL/ARCHIVE**: physical-consequence attention is useful, but the present mechanism does not support an ICLR-main-target claim.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. The v4 repository contains a paper-specific attention-budget benchmark, implemented baselines, ablations, stress sweeps, figures, negative cases, and reproducibility files. The evidence shows that token salience is a poor control allocation rule, but it does not show that the proposed consequence-budgeting mechanism is decisively better than strong risk-aware or active-perception baselines.

2 RESEARCH QUESTION

The intended claim was:

> Allocate perception and action attention by physical consequence rather than token salience.

This is a meaningful embodied-learning distinction. A visually salient object may be harmless, while a visually dull contact edge, hidden support surface, fragile obstacle, or cable snag may decide the task. A useful policy must trade off critical-event recall, wasted attention, compute cost, latency, and safety.

3 HOSTILE PRIOR WORK

The local hostile set includes embodied foundation models, robot foundation-model surveys, physical-risk control, embodied VLA systems, and work warning that attention weights often capture correlations rather than causal task structure. Therefore, the paper cannot claim novelty from generic attention, token pruning, active perception, uncertainty, or compute adaptation. The benchmark tests whether physical consequence changes budget allocation and downstream robot behavior.

4 Benchmark

The v4 benchmark simulates four settings:

- **cluttered pick-and-place** with distractors and hidden support surfaces.
- **mobile manipulation** around fragile obstacles.
- **tool-use sequencing** with low-salience but high-consequence contact points.
- **deformable cable routing** with latent snags and occlusion.

Each episode samples candidate perception/action tokens with visual salience, language salience, uncertainty, physical consequence, hazard level, action-criticality, compute cost, and a true critical-token label. A method spends a fixed attention budget before acting. Splits include balanced nominal, visual distractor shift, physical consequence shift, tight compute budget, and combined hard shift.

5 Methods

We compare:

- **Uniform attention budget:** spends budget without learned prioritization.
- **Token-salience transformer:** ranks by visual and language salience.
- **Uncertainty attention:** prioritizes uncertain tokens.
- **Active perception value:** blends uncertainty and estimated action value.
- **Compute-adaptive policy:** normalizes attention value by compute cost.
- **Risk-aware attention:** emphasizes hazard and uncertainty.
- **Embodied consequence budgeting:** the proposed physical-consequence, action-critical, hazard-aware budget.
- **Oracle consequence attention:** hidden true physical score.

6 Main Results

Table 1 reports the combined hard-shift split. The proposed method is much better than token salience and uncertainty-only attention, which waste budget on distractors or uncertain but harmless tokens. However, the margin over risk-aware attention is not decisive. Risk-aware attention reaches 0.676 success, while consequence budgeting reaches 0.693, for a paired success difference of 0.01701 ± 0.04077.

Table 1: Combined hard-shift results over seven seeds. Confidence intervals are 95% intervals over seed means. Lower wasted attention, latency violation, safety violation, and regret are better.

Method	Success	Recall	Wasted	Safety	Regret
Uniform	0.241 ± 0.021	0.175	0.677	0.317	0.522
Token salience	0.073 ± 0.013	0.024	0.952	0.417	0.739
Uncertainty	0.205 ± 0.012	0.149	0.696	0.325	0.534
Active perception	0.629 ± 0.031	0.505	0.118	0.195	0.152
Compute adaptive	0.423 ± 0.010	0.317	0.473	0.263	0.363
Risk aware	0.676 ± 0.021	0.519	0.101	$\mathbf{0.158}$	0.139
Consequence budget	$\mathbf{0.693 \pm 0.029}$	0.562	0.062	0.163	0.105
Oracle consequence	0.713 ± 0.026	$\mathbf{0.567}$	$\mathbf{0.053}$	0.145	$\mathbf{0.098}$

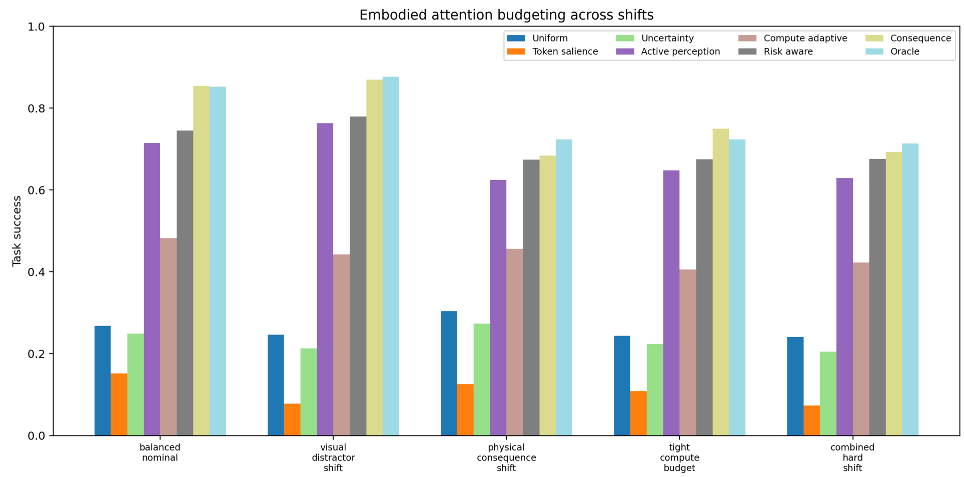

Figure 1: Task success across attention-budget shifts. Consequence budgeting is useful but not decisively ahead of risk-aware attention.

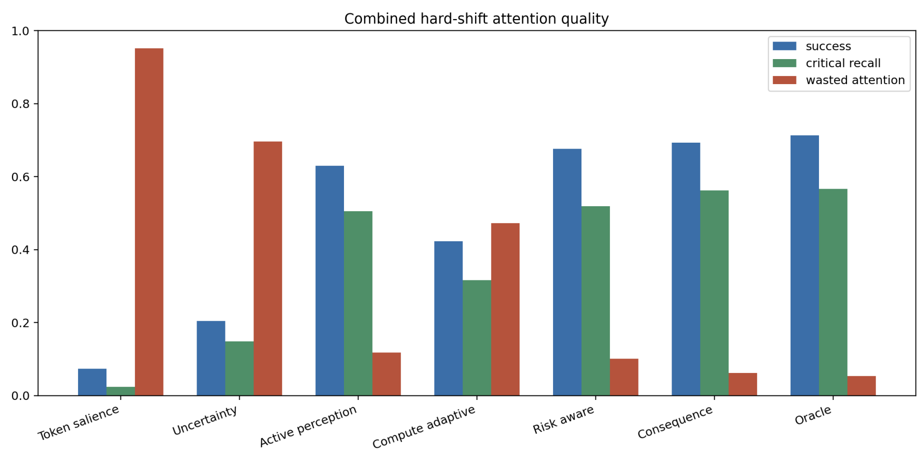

Figure 2: Combined hard-shift success, critical-event recall, and wasted attention. Token salience wastes budget under visual distractors.

7 Ablations

The ablations do not validate the proposed mechanism. Removing the physical consequence estimator collapses performance, so consequence information matters. But removing the action-critical contact map improves task success to 0.705, and removing the safety term also slightly improves success while worsening safety. This means the full design is not uniquely justified.

8 Stress Tests

Stress sweeps vary visual distractor load, occlusion, compute-budget tightening, latency penalty, hidden physical consequence, and combined stress. At maximum combined stress, risk-aware attention reaches 0.618 task success, while consequence budgeting reaches 0.602. The proposed method has better safety at that point, but the success reversal weakens the submission claim.

Table 2: Embodied attention-budget ablations on combined hard shift.

Ablation	Success	Recall	Wasted	Safety
Full consequence budgeting	0.693 ± 0.029	$\mathbf{0.562}$	0.062	0.163
Minus physical consequence	0.232 ± 0.027	0.165	0.673	0.342
Minus action-critical map	$\mathbf{0.705 \pm 0.041}$	0.533	0.091	$\mathbf{0.154}$
Minus compute-latency constraint	0.689 ± 0.023	0.557	$\mathbf{0.049}$	0.156
Minus safety hazard term	0.695 ± 0.022	0.545	0.075	0.192
Token-salience only	0.063 ± 0.010	0.024	0.952	0.417
Uncertainty only	0.227 ± 0.021	0.149	0.696	0.332

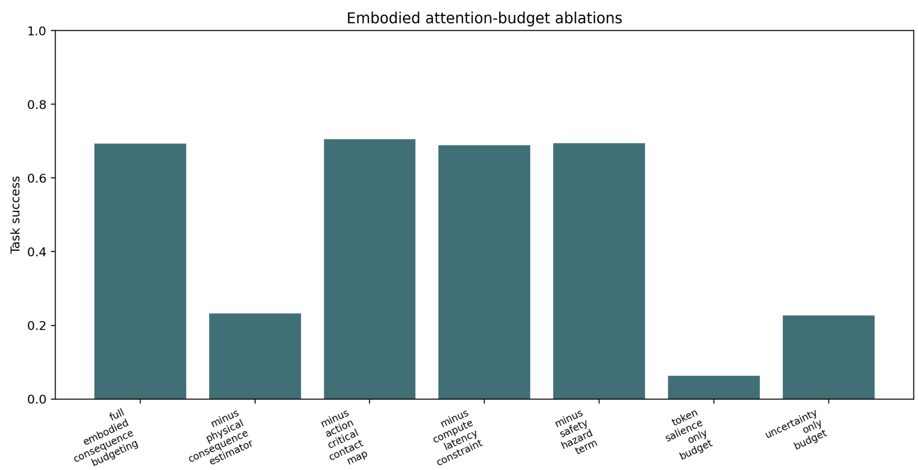

Figure 3: Ablation success. A core ablation exceeds the full method, so the proposed design fails a mechanism gate.

9 Why This Is Not Submission-Ready

The project fails the ICLR-main readiness gate for four reasons.

- **Non-decisive baseline margin.** The paired success gain over risk-aware attention is only 0.01701 ± 0.04077.
- **Ablation contradiction.** Removing the action-critical contact map improves success.
- **Stress reversal.** Risk-aware attention beats the proposed method at maximum combined stress.
- **Evidence ceiling.** The benchmark is deterministic and reproducible but local; there is no robot hardware, accepted external benchmark, or implemented foundation-model attention module.

10 Reproducibility

Run the benchmark from the repository root:

```
python src\run_experiment.py
```

The script writes rollout CSVs, seed-level metrics, pairwise statistics, ablations, stress sweeps, negative cases, summary text, and the figures used in this manuscript. The canonical local PDF is `C:/Users/wangz/Downloads/89.pdf`.

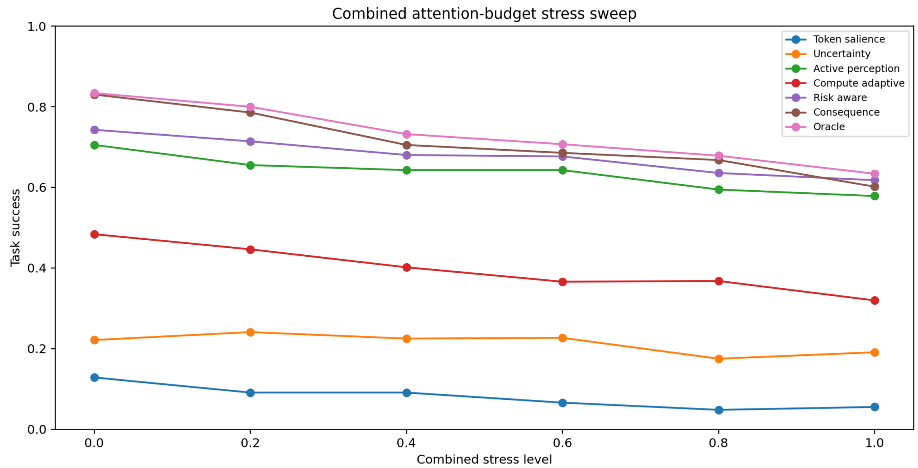

Figure 4: Combined stress sweep. Risk-aware attention slightly exceeds consequence budgeting at maximum stress.

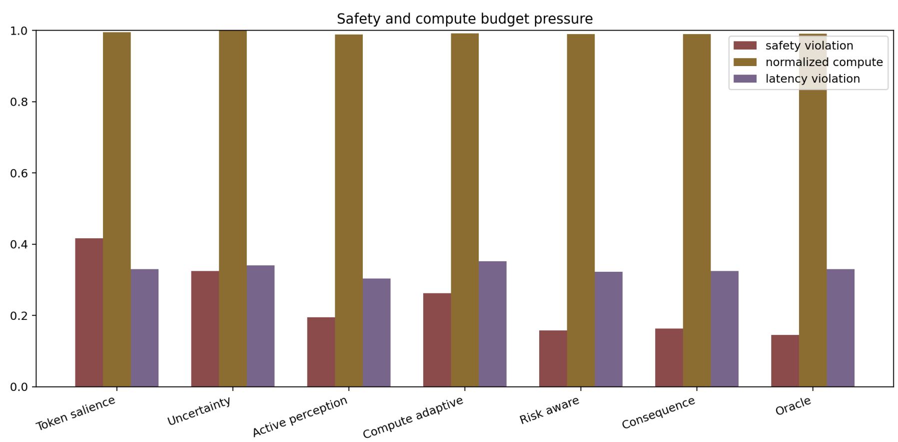

Figure 5: Safety and compute pressure. Consequence budgeting is competitive, but not a decisive Pareto improvement.

11 Conclusion

Embodied attention should not be reduced to token salience. The v4 benchmark demonstrates that physical consequence and risk matter. But the proposed mechanism is not yet a submission-ready answer: it is statistically too close to risk-aware attention, contradicted by ablations, and not validated externally. The honest terminal action is **KILL/ARCHIVE**.